Flask Framework — Detailed Theory & Concepts

What is Flask?

Flask is a lightweight micro web framework written in Python. It is used to build web applications, APIs, dashboards, and backend systems.

Why Flask is Called a Micro Framework

Flask only provides the essential components required for web development, such as routing, request handling, template rendering, and a development server.

Core Philosophy of Flask

1. Simplicity — Easy to learn and use.
2. Flexibility — Developers can choose their own tools and architecture.
3. Extensibility — Additional features can be added using extensions.

Flask Architecture

Flask follows the client-server model. The browser sends an HTTP request, Flask processes it, and returns a response.

WSGI Concept

WSGI stands for Web Server Gateway Interface. It acts as a bridge between Python applications and web servers.

Basic Flask Application

```
from flask import Flask

app = Flask(__name__)

@app.route('/')
def home():
    return "Hello Flask"

if __name__ == '__main__':
    app.run(debug=True)
```

Routing

Routing connects URLs with Python functions using the route decorator.

Dynamic Routing

Flask supports variables inside URLs for dynamic routing.

HTTP Methods

Flask supports GET, POST, PUT, and DELETE methods for handling HTTP requests.

Request Object

The request object stores incoming request data such as form data, headers, JSON data, and cookies.

Response Object

Flask view functions can return strings, HTML pages, templates, JSON data, or redirects.

Templates and Jinja2

Flask uses the Jinja2 template engine for generating dynamic HTML pages.

Static Files

CSS, JavaScript, and images are stored in the static folder.

Debug Mode

Debug mode automatically reloads the server and displays detailed errors during development.

Sessions and Cookies

Sessions store temporary user-specific data while cookies are stored inside the browser.

Flask Extensions

Popular Flask extensions include Flask-SQLAlchemy, Flask-Login, Flask-WTF, and Flask-Mail.

Database Integration

Flask supports databases such as SQLite, MySQL, PostgreSQL, and MongoDB.

Blueprints

Blueprints help organize large Flask applications into smaller modules.

REST APIs

Flask is widely used to build REST APIs because it is lightweight and easy to work with JSON.

Advantages of Flask

- Easy to learn
- Flexible architecture
- Lightweight framework
- Excellent for APIs and ML deployment

Disadvantages of Flask

- Manual setup required for large applications
- Fewer built-in tools compared to Django

Production Deployment

Flask applications are commonly deployed using Gunicorn and Nginx.

Conclusion

Flask is one of the best frameworks for beginners and backend developers. It is lightweight, flexible, modular, and widely used for APIs and web applications.